Worksheet 4D: Transcendental Functions via Series

AIML Foundations Mathematics
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What This Worksheet Is About *The functions e^x, $\sin(x)$, $\cos(x)$, and $\ln(x)$ aren't polynomials — they're* ***transcendental functions****. But here's the beautiful secret: They can be written as* ***infinite polynomials*** *(power series)! Once we write them as series, we can differentiate and integrate term by term — discovering their derivatives and integrals through pure algebra!*

Part A: The Exponential Function e^x (14 problems)

The Series Definition:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

where $n! = n \times (n-1) \times \cdots \times 2 \times 1$ (factorial)

Understanding the Series:

1. Write out the first 6 terms of the series for e^x: $e^x = 1 + x + \frac{x^2}{2} +$ ____________ $+$ ____________ $+$ ____________ $+ \cdots$
2. Calculate e^0 using the series (plug in $x = 0$): $e^0 = 1 + 0 + 0 + 0 + \cdots =$ ____________
3. Approximate e^1 using the first 6 terms: $e^1 \approx 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} =$ ____________ (Actual value: $e \approx 2.71828...$)
4. Approximate $e^{0.1}$ using the first 4 terms: $e^{0.1} \approx 1 + 0.1 + \frac{0.01}{2} + \frac{0.001}{6} =$ ____________

Differentiating e^x Term by Term:

5. Differentiate each term of the series:

Term	Derivative
1	0
x	1
$\frac{x^2}{2!} = \frac{x^2}{2}$	$\frac{2x}{2} = x$
$\frac{x^3}{3!} = \frac{x^3}{6}$	____________
$\frac{x^4}{4!} = \frac{x^4}{24}$	____________
$\frac{x^5}{5!}$	____________

6. Write out the derivative series: $\frac{d}{dx}e^x = 0+1+x+$ ____________________ $+$ ____________________ $+$ ____________________ $+\cdots$
7. Compare to the original series. What do you notice?

$$\boxed{\frac{d}{dx}e^x = e^x}$$

The exponential function is its own derivative!

8. Since $\frac{d}{dx}e^x = e^x$, what is $\int e^x\, dx$?
9. Verify by integrating the series term by term: $\int e^x dx = \int(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots)dx$ $= x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots + C = (1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots) - 1 + C = e^x + C'$ **With Chain Rule:**
10. Using the chain rule, find $\frac{d}{dx}e^{2x}$.
11. Find $\frac{d}{dx}e^{x^2}$.
12. Find $\int e^{3x}dx$.
13. Find $\int 2xe^{x^2}dx$ using substitution.
14. The function e^x satisfies $\frac{dy}{dx} = y$. Explain why this makes e^x special for modeling growth.

Part B: Sine and Cosine via Series (18 problems)

The Series Definitions:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

Note: x must be in **radians**!

Understanding the Series:

15. Notice: $\sin(x)$ has only _________________________ powers of x (1, 3, 5, 7, ...)
16. Notice: $\cos(x)$ has only _________________________ powers of x (0, 2, 4, 6, ...)
17. Calculate $\sin(0)$ using the series: $\sin(0) = 0 - 0 + 0 - \cdots =$ ____________________
18. Calculate $\cos(0)$ using the series: $\cos(0) = 1 - 0 + 0 - \cdots =$ ____________________
19. Approximate $\sin(0.1)$ using first 3 non-zero terms: $\sin(0.1) \approx 0.1 - \frac{0.001}{6} + \frac{0.00001}{120} =$ ____________________ (Actual: $\sin(0.1) = 0.0998334...$)
20. Approximate $\cos(0.1)$ using first 3 non-zero terms.

Differentiating Sine Term by Term:

21. Differentiate each term of $\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \cdots$

Term	Derivative
x	1
$-\frac{x^3}{6}$	$-\frac{3x^2}{6} = -\frac{x^2}{2}$
$\frac{x^5}{120}$	________________
$-\frac{x^7}{5040}$	________________

22. Write the derivative series: $\frac{d}{dx}\sin(x) = 1 - \frac{x^2}{2} +$ ________________ - ________________ $+ \cdots$

23. Compare to the cosine series. What do you notice?

$$\boxed{\frac{d}{dx}\sin(x) = \cos(x)}$$

Differentiating Cosine Term by Term:

24. Differentiate $\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots$

Term	Derivative
1	0
$-\frac{x^2}{2}$	$-x$
$\frac{x^4}{24}$	________________
$-\frac{x^6}{720}$	________________

25. Write the derivative series: $\frac{d}{dx}\cos(x) = 0 - x +$ ________________ - ________________ $+ \cdots$

26. Factor out -1: $= -(x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots)$ What function is this?

$$\boxed{\frac{d}{dx}\cos(x) = -\sin(x)}$$

The Beautiful Cycle:

27. Complete the derivative cycle:

$$\sin(x) \xrightarrow{\frac{d}{dx}} \cos(x) \xrightarrow{\frac{d}{dx}} -\sin(x) \xrightarrow{\frac{d}{dx}} -\cos(x) \xrightarrow{\frac{d}{dx}} \_\_\_\_\_\_\_\_$$

28. After how many derivatives does sine return to itself?

29. What is $\frac{d^{100}}{dx^{100}}\sin(x)$? (Hint: $100 \div 4 = ?$)

Integrals (Reversing):

30. Since $\frac{d}{dx}\sin(x) = \cos(x)$, what is $\int \cos(x)dx$?

31. Since $\frac{d}{dx}\cos(x) = -\sin(x)$, what is $\int \sin(x)dx$?

32. Verify $\int \sin(x)dx = -\cos(x) + C$ by differentiating.

Part C: The Natural Logarithm $\ln(x)$ (12 problems)

Finding the Derivative:

The logarithm is defined as the inverse of e^x. If $y = \ln(x)$, then $e^y = x$.

33. Differentiate $e^y = x$ implicitly: $e^y \cdot \frac{dy}{dx} = 1$ $\frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$

$$\boxed{\frac{d}{dx}\ln(x) = \frac{1}{x}}$$

34. What is $\int \frac{1}{x} dx$?
35. Remember from 4A: We couldn't integrate x^{-1} using the power rule because it gave $\frac{x^0}{0}$. Now we know: $\int x^{-1} dx =$ ______________________________

The Series for $\ln(1+x)$:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad (\text{for } -1 < x \leq 1)$$

36. Differentiate term by term: $\frac{d}{dx}\ln(1+x) = 1 - x + x^2 - x^3 + \cdots$
37. The series $1 - x + x^2 - x^3 + \cdots$ is a geometric series with first term 1 and ratio $-x$. Sum $= \frac{1}{1-(-x)} = \frac{1}{1+x}$ Verify: $\frac{d}{dx}\ln(1+x) = \frac{1}{1+x}$ (chain rule!) **With Chain Rule:**
38. $\frac{d}{dx}\ln(2x)$
39. $\frac{d}{dx}\ln(x^2)$
40. $\frac{d}{dx}\ln(x^2+1)$
41. $\int \frac{2x}{x^2+1} dx$ (substitution: let $u = x^2 + 1$)
42. $\int \frac{1}{2x+3} dx$ **Connection:**
43. e^x and $\ln(x)$ are inverse functions. - $\frac{d}{dx}e^x = e^x$ - $\frac{d}{dx}\ln(x) = \frac{1}{x}$ At $x = 1$: $\frac{d}{dx}e^x\Big|_{x=0} = e^0 = 1$ and $\frac{d}{dx}\ln(x)\Big|_{x=1} = 1$ The slopes are ______________________________ at corresponding points!
44. For inverse functions f and f^{-1}: $\frac{d}{dx}f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$ Verify this for e^x and $\ln(x)$.

Part D: Integration by Parts with Transcendentals (10 problems)

Now we can use by parts with these functions!

$$\int u\,dv = uv - \int v\,du$$

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45. $\int xe^x dx$ Let $u = x$, $dv = e^x dx$ Then $du = dx$, $v = e^x = xe^x - \int e^x dx = xe^x - e^x + C = e^x(x-1) + C$
46. Verify by differentiating $e^x(x-1)$.
47. $\int x\cos(x)dx$ Let $u = x$, $dv = \cos(x)dx$
48. $\int x\sin(x)dx$
49. $\int x^2 e^x dx$ (need by parts twice!)
50. $\int e^x \sin(x)dx$ (a famous one — try it!) *Hint: Use by parts twice, you'll get the original integral on the right side. Solve for it!*
51. $\int \ln(x)dx$ Let $u = \ln(x)$, $dv = dx$ Then $du = \frac{1}{x}dx$, $v = x = x\ln(x) - \int x \cdot \frac{1}{x}dx = x\ln(x) - x + C$
52. Verify by differentiating $x\ln(x) - x$.
53. $\int x\ln(x)dx$
54. $\int (\ln x)^2 dx$ (by parts twice)

Part E: Euler's Formula — The Most Beautiful Equation (8 problems)

The Connection:

Using the series: - $e^{ix} = 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \cdots$

Since $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^5 = i$, ...

55. Expand the first 6 terms of e^{ix}: $e^{ix} = 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \frac{ix^5}{5!} - \cdots$
56. Separate into real and imaginary parts: Real: $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots =$ ______________________________ Imaginary: $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots =$ ______________________________
57. Therefore:

    $$\boxed{e^{ix} = \cos(x) + i\sin(x)}$$

    This is **Euler's Formula**!
58. Plug in $x = \pi$: $e^{i\pi} = \cos(\pi) + i\sin(\pi) =$ ____________________ + ____________________ $= -1$

    $$\boxed{e^{i\pi} + 1 = 0}$$

    This is "Euler's Identity" — often called the most beautiful equation in mathematics!
59. Using Euler's formula, show that: $\cos(x) = \frac{e^{ix}+e^{-ix}}{2}$ $\sin(x) = \frac{e^{ix}-e^{-ix}}{2i}$
60. Now we can see WHY $\frac{d}{dx}\sin(x) = \cos(x)$: Differentiate $\sin(x) = \frac{e^{ix}-e^{-ix}}{2i}$ using chain rule on each exponential.
61. These connections show that e, π, i, sin, and cos are all deeply related. In your own words, why is this surprising/beautiful?

62. The formula $e^{ix} = \cos(x) + i\sin(x)$ means exponential growth in the imaginary direction produces ______________________________.

Part F: Summary — The Big Picture (6 problems)

63. Complete the derivative table:

Function	Derivative
e^x	
$\ln(x)$	
$\sin(x)$	
$\cos(x)$	

64. Complete the integral table:

Function	Integral
e^x	
$\frac{1}{x}$	
$\sin(x)$	
$\cos(x)$	

65. We derived all of these using ______________________________ expansions!

66. The key insight: Transcendental functions can be written as ______________________________ polynomials, and we know how to differentiate polynomials term by term.

67. Why can't we integrate e^{-x^2} in closed form? *The series would be $1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots$, and integrating gives a series with no nice closed form.*

68. This is why numerical methods matter in ML and statistics — the Gaussian/Normal distribution involves e^{-x^2}!